Chapter 9: Maxwell's Equations

The following subjects will be discussed

- Faraday's Law of Induction.
- Maxwell's Equations.
- Displacement Current Density
- Time-harmonic Fields.

Case (1): Transformational EMF

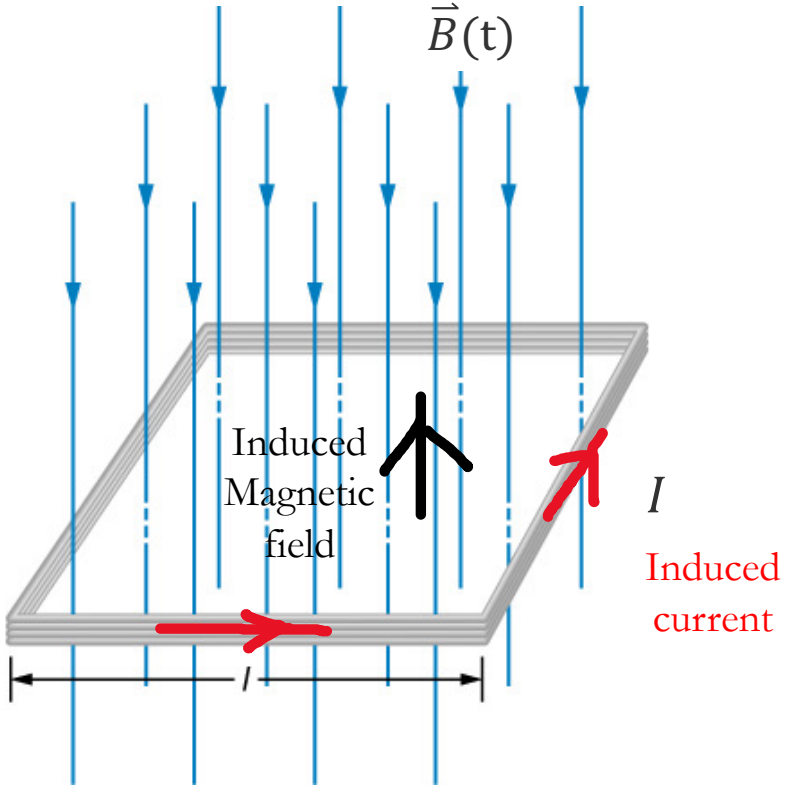
A conducting wire (circuit) with N turns is stationary inside a region of uniform and time-varying magnetic field. There is a change in flux linking the circuit with time.

$$\phi(t) = N \int \vec{B}(t) \cdot \overrightarrow{dA}$$

$$Emf = -\frac{d\phi\,(t)}{dt}$$

$$Emf = -N \int \frac{d\vec{B}(t)}{dt} \cdot \overrightarrow{dA}$$

The magnitude of the EMF can be found using the above equation (without the minus sign). The polarity of the EMF (or direction of induced current in the circuit) can be found using Lenz's law.



Case (2): Motional EMF

A conducting rod is moved inside a region of constant magnetic field.

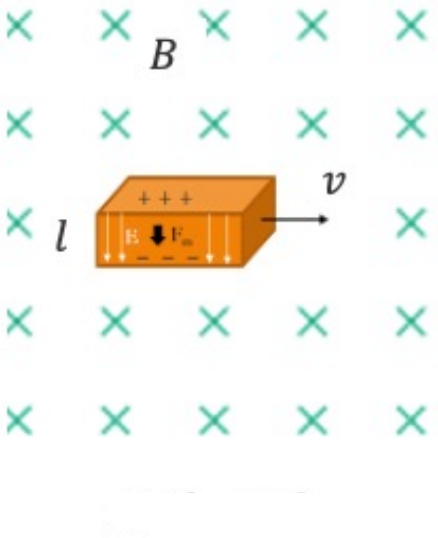
Change in flux is not involved here. However, magnetic and electric forces are applied to the charges within the conductor

At equilibrium,

$$\vec{F}_m = \vec{F}_e$$

$$q\vec{E} = q(\vec{v}\times\vec{B})$$

$$Emf = \int \vec{E}\cdot\overrightarrow{dl} = \int (\vec{v}\times\vec{B})\cdot\overrightarrow{dl}$$



For constant velocity and uniform magnetic field,

$$Emf = vBl$$

Case (2): Motional EMF

A conducting rod is moved inside a region of constant magnetic field.

Change in flux is not involved here. However, magnetic and electric forces are applied to the charges within the conductor

For constant angular velocity and uniform magnetic field,

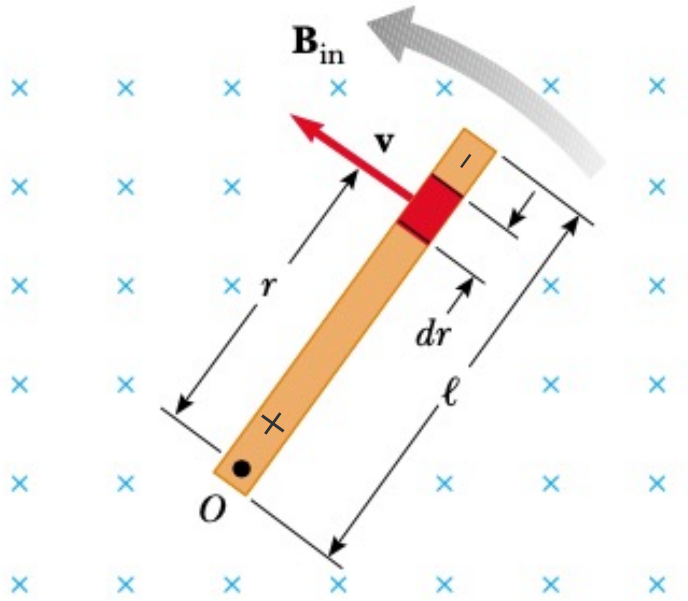
$$Emf = vBl$$

$$d(Emf) = vB(dr)$$

$$v = \omega r$$

$$Emf = \int_0^l \omega B\, r\, dr$$

$$Emf = \frac{1}{2}\omega B l^2$$



Case (3): Transformational and Motional EMF

A part of the circuit is moved inside a region of time-varying magnetic field.

This case can be solved by adding the change in flux linking the circuit, and motional consideration.

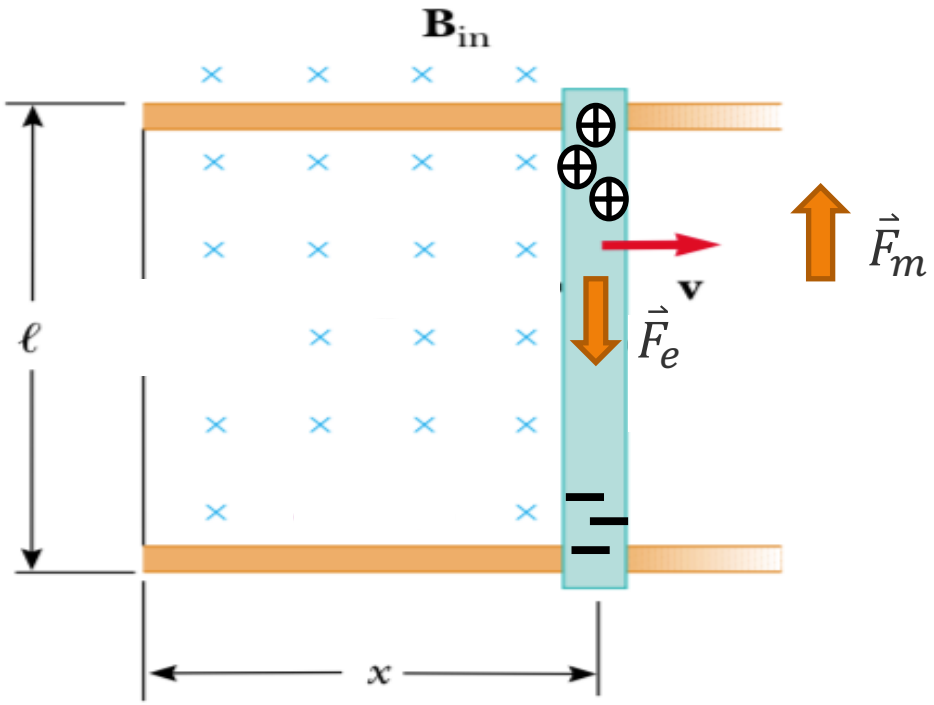
$$Emf = -\int \frac{d\vec{B}(t)}{dt} \cdot \overrightarrow{dA} + \int (\vec{v} \times \vec{B}) \cdot \overrightarrow{dl}$$

In this case

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} + \vec{\nabla} \times (\vec{v} \times \vec{B})$$

Note that it is also possible to write the total time-varying flux, taking into account the moving part, and then using

$$Emf = -\frac{d\phi\,(t)}{dt}$$



Current direction is such that it opposes the change in linkage!

Lenz's Law

Examples

(1) Consider a square coil of 200 turns and size 18 cm x 18 cm is placed in a plane normal to a time-varying magnetic field given by $B(t) = 0.625\,t\ (T)$.
Find the induced emf.

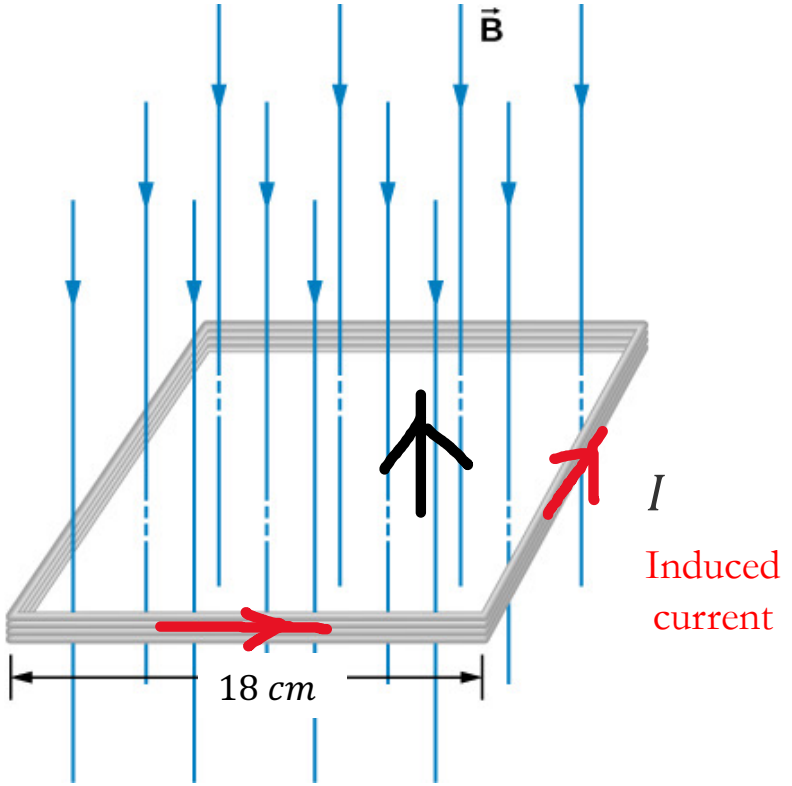
$$Emf = -\frac{d\phi}{dt}$$

$$\phi = \int \vec{B} \cdot \overrightarrow{dA}$$

$$\phi = NBA = (200)(0.625\,t)\,(0.18)(0.18)$$

$$Emf = -(200)(0.625)(0.18)(0.18)$$

$$Emf = -4.05\ V$$



Examples

(2) A conducting ring of radius r and total resistance R is kept fixed on the xy-plane. Starting at time t=0, a uniform magnetic field $\vec{B}(t) = \alpha t(\vec{a}_y + 2\vec{a}_z)$, where α is a positive constant, is applied. Find the magnitude and direction of the current induced in the ring.

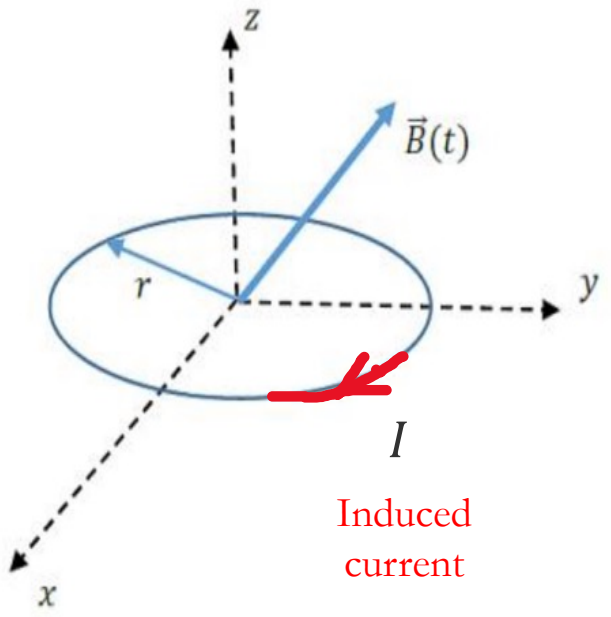
$$\phi = \int \vec{B} \cdot \overrightarrow{dA}$$

$$Emf = -\frac{d\phi}{dt}$$

$$Emf = -\frac{d\phi}{dt} = -\int \frac{d\vec{B}}{dt} \cdot \overrightarrow{dA}$$

$$Emf = -2\alpha(\pi r^2)$$

$$I = \frac{Emf}{R} = \frac{2\pi\alpha r^2}{R}$$



Examples

(3) Find expressions for the induced current and the forces acting on the loop while it is being pulled out of the uniform and constant magnetic field region with constant velocity.

$$\phi = \int \vec{B} \cdot \overrightarrow{dA} = BA = BLx$$

$$Emf = -\frac{d\phi}{dt} = BL\frac{dx}{dt} = BLv$$

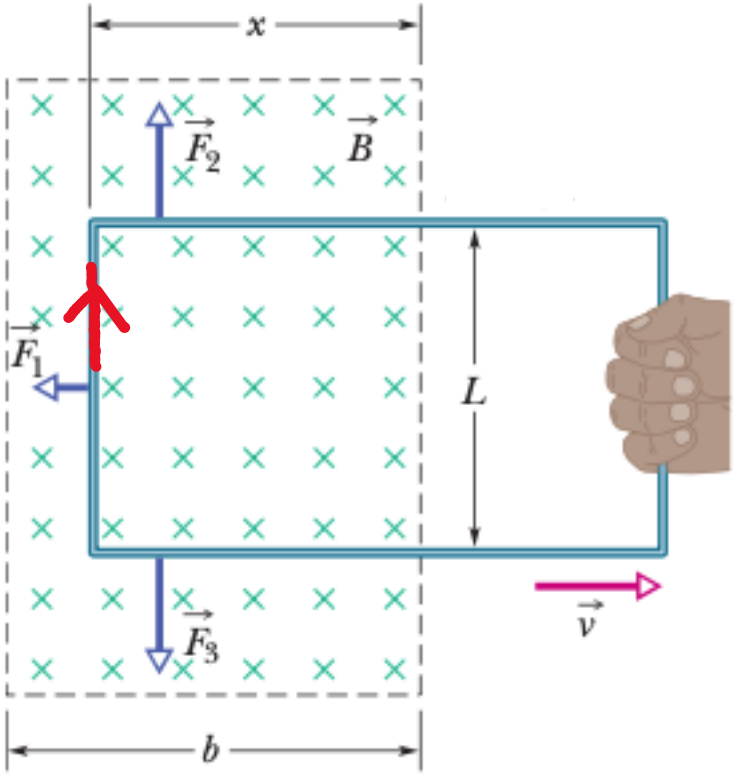
$$I = \frac{Emf}{R}$$

$$\vec{F}_m = I\,(\vec{l}_{ab} \times \vec{B})$$

$$F_1 = I\,B\,L = \frac{Emf}{R}BL = \frac{(BL)^2\,v}{R}$$

$$F_2 = I\,B\,x = \frac{Emf}{R}BL = \frac{B^2\,Lxv}{R}$$

$$F_3 = -F_2$$



Examples

(4) A rod of length L is forced to move at constant speed v along horizontal rails. The rod, rails, and connecting strip at the right form a conducting loop. The rod has resistance R and the rest of the loop has negligible resistance. A long straight wire at distance a from the top of loop carries a current i. This wire generates a nonuniform magnetic field through the loop. Find the induced current in the rod.

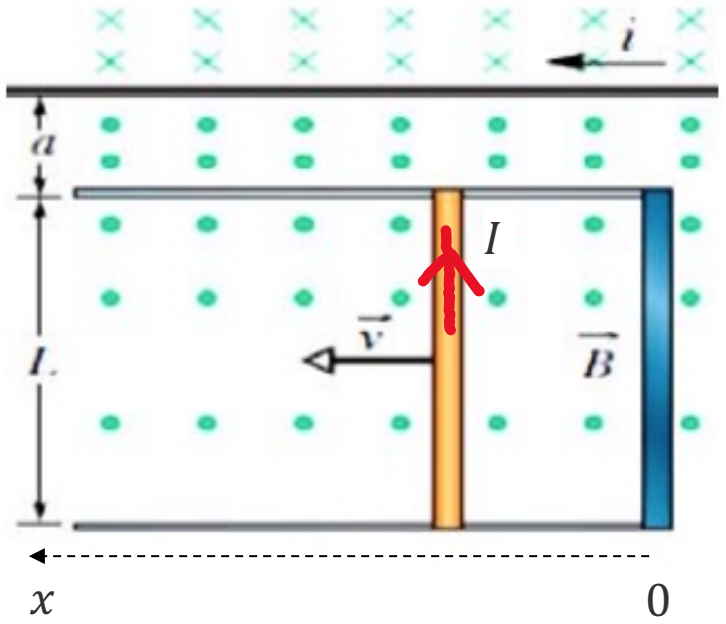
$$B = \frac{\mu_o i}{2\pi y}$$

$$\phi = \int \vec{B} \cdot \overrightarrow{dA} \quad \rightarrow \quad \phi = \frac{\mu_o i\, x}{2\pi} \ln\left(\frac{a+L}{a}\right)$$

$$Emf = -\frac{d\phi}{dt} = -\frac{\mu_o i}{2\pi} \ln\left(\frac{a+L}{a}\right)\frac{dx}{dt}$$

$$Emf = \frac{\mu_o i}{2\pi} \ln\left(\frac{a+L}{a}\right) v$$

$$I = \frac{Emf}{R}$$

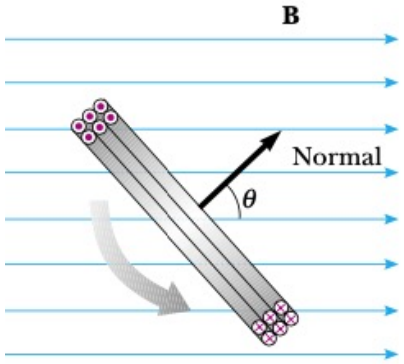


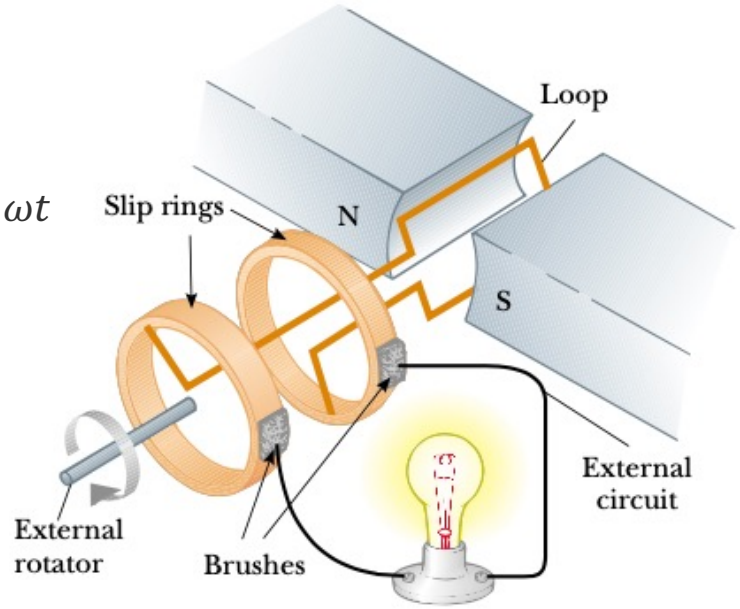
The Electric Generator

A loop with N turns is placed inside a magnetic field. This magnetic field is generated by permanent magnets. The loop is rotated by an external mechanism. An emf is generated due to change in flux. The terminals of the loop are connected to a load. The polarity is fixed.

$$\phi = \int \vec{B} \cdot \overrightarrow{dA}$$

$$\phi = NBA\cos\theta = NBA\cos\omega t$$





$$Emf = -\frac{d\phi}{dt} = NBA\omega\sin\omega t$$

The standard frequencies are 50 Hz and 60 Hz

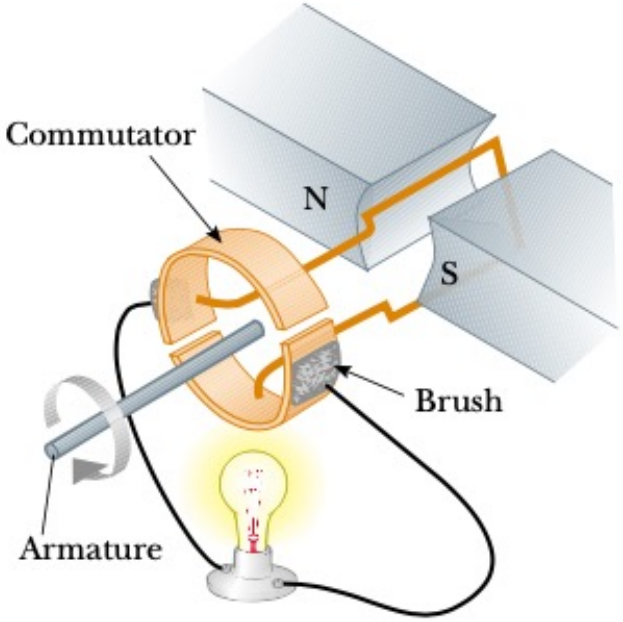
Example: Area=0.09 m^2, turns=8, magnetic field=0.5 T, frequency=60 Hz and loop resistance=12 Ω.

$$Emf = (8)(0.09)(0.5)(2\pi 60)\sin\omega t = 136\sin\omega t \quad (V)$$

$$I_{max} = 136/12 = 11.3 \quad (A)$$

The Electric Generator

A loop with N turns is placed inside a magnetic field. This magnetic field is generated by permanent magnets. The loop is rotated by an external mechanism. An emf is generated due to change in flux. The terminals of the loop are connected to a load. The polarity is reversed.



DC generator



The Displacement Current Density

An inconsistency can be found between the original Ampere's equation and the continuity equation (Maxwell's finding).

Ampere's Law in differential form. $(\vec{\nabla} \times \vec{H}) = \vec{J}$

Continuity equation $\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$

Taking the divergence of both sides in Ampere's law,

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} \quad \rightarrow \quad \vec{\nabla} \cdot \vec{J} = 0$$

Inconsistent with continuity equation.

This results contradicts with the continuity equation. To remedy the situation, Maxwell added the appropriate term to the above equation, as

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} \quad \rightarrow \quad \vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \frac{\partial \vec{\nabla} \cdot \vec{D}}{\partial t} = \vec{\nabla} \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

Removing the divergence from both sides,

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Modified Ampere's law (Maxwell's equation)

Conduction current density (A/m^2)

$$\vec{J}_c = \sigma \vec{E}$$

Displacement current density (A/m^2)

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \varepsilon \frac{\partial \vec{E}}{\partial t}$$

The displacement current exists solely due to electric field time variation. It vanishes for static fields.

It depends on the dielectric properties of the medium.

The Displacement Current Density

Consider a parallel plate capacitor with dimension d much smaller than other structure dimensions. The capacitance is given by, $C = \varepsilon_o \varepsilon_r \frac{A}{d}$

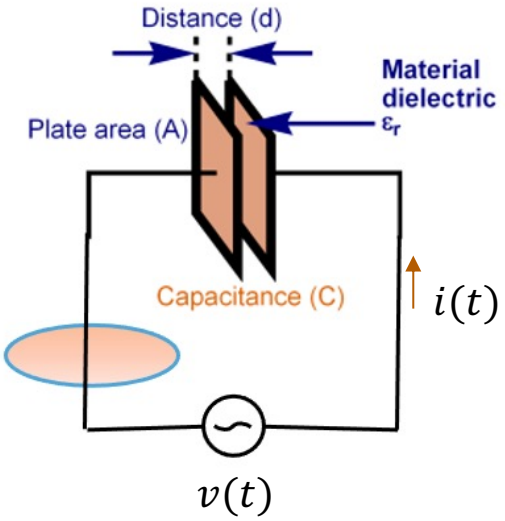
Let's calculate the magnitude of the displacement current generated across the dielectric material filling the capacitor.

$$J_d = \frac{\partial D}{\partial t} = \varepsilon_o \varepsilon_r \frac{\partial E}{\partial t} = \frac{\varepsilon_o \varepsilon_r}{d} \frac{\partial v(t)}{\partial t}$$

$$i(t) = I_d = \int_S J_d \, aS = \varepsilon_o \varepsilon_r \frac{A}{d} \frac{\partial v(t)}{\partial t}$$

$$i(t) = C \frac{\partial v(t)}{\partial t}$$

This is the equation used in circuit analysis, which explains the current in capacitors.



Maxwell's Equations

EM Fields Equations

Any electromagnetic field must satisfy these equations.

Differential Form

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$$

$$\vec{\nabla}\cdot\vec{D} = \rho_v$$

$$\vec{\nabla}\cdot\vec{B} = 0$$

Integral Form

$$\oint \vec{E}\cdot\overrightarrow{dl} = \frac{d}{dt}\int \vec{B}(t)\cdot\overrightarrow{dS}$$

$$\oint \vec{H}\cdot\overrightarrow{dl} = \int \vec{J}\cdot\overrightarrow{dS} + \frac{d}{dt}\int \vec{D}(t)\cdot\overrightarrow{dS}$$

$$\oint \vec{D}\cdot\overrightarrow{dS} = Q$$

$$\oint \vec{B}\cdot\overrightarrow{dS} = 0$$

Continuity Equation

$$\vec{\nabla}\cdot\vec{J} + \frac{\partial}{\partial t}\rho_v = 0$$

EM Boundary Conditions

$$E_{1t} = E_{2t} \qquad D_{1n} = D_{2n} + \rho_s$$

$$\vec{H}_1\times\vec{a}_n - \vec{H}_2\times\vec{a}_n = \vec{K} \qquad B_{1n} = B_{2n}$$

Constitutive Relations (material)

$$\vec{J} = \sigma\vec{E} \qquad \vec{B} = \mu\vec{H} \qquad \vec{D} = \varepsilon\vec{E}$$

EM Energy Density

$$w_E = \frac{\varepsilon}{2}\left|\vec{E}\right|^2 \qquad w_m = \frac{\varepsilon}{2}\left|\vec{H}\right|^2$$

Examples

(1) The electric field distribution in free space is $\vec{E}(x,t) = 20\cos(\omega t - 50x)\,\vec{a}_y$ (V/m). Find the displacement current density and the corresponding magnetic field.

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \varepsilon_o \frac{\partial \vec{E}}{\partial t} = -20\varepsilon_o \omega \sin(\omega t - 50x)\;\vec{a}_y \quad (\mathrm{A/m^2})$$

Any electromagnetic field must satisfy Maxwell's equations.

$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\frac{dE_y}{dx}\vec{a}_z = -\mu_o \frac{\partial \vec{H}}{\partial t}$$

$$\frac{\partial \vec{H}}{\partial t} = -\frac{1}{\mu_o}(20\times 50)\sin(\omega t - 50x)\,\vec{a}_z$$

$$\vec{H}(x,t) = \frac{1000}{\mu_o \omega}\cos(\omega t - 50x)\,\vec{a}_z \quad (\mathrm{A/m})$$

Time-harmonic Maxwell's Equations

Consider the special case where the time variation of the fields/charges/currents is harmonic in time (sinusoidal).

For generality and convenience, the harmonic behavior is represented in term of exponentials.

$e^{j\omega t}$ Sinusoidal, single frequency (ω)

In this case, the fields can be expressed as,

$$\vec{E}(r,t) = \vec{E}(r)\, e^{j\omega t}$$

$$\vec{H}(r,t) = \vec{H}(r)\, e^{j\omega t}$$

$$\vec{D}(r,t) = \vec{D}(r)\, e^{j\omega t}$$

$$\vec{B}(r,t) = \vec{B}(r)\, e^{j\omega t}$$

$$\vec{J}(r,t) = \vec{J}(r)\, e^{j\omega t}$$

$$\rho_v(r,t) = \rho_v(r)\, e^{j\omega t}$$

r represent any space coordinate system variables(not necessarily spherical)

Maxwell's equations can be modified accordingly by evaluating the time derivative. Here is an example.

Starting with

$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla}\times\left(\vec{E}(r)\, e^{j\omega t}\right) = -\frac{\partial}{\partial t}\left(\vec{B}(r)\, e^{j\omega t}\right)$$

$$e^{j\omega t}\,\vec{\nabla}\times\vec{E}(r) = -\vec{B}(r)\frac{\partial}{\partial t}\left(e^{j\omega t}\right)$$

$$e^{j\omega t}\,\vec{\nabla}\times\vec{E}(r) = -\vec{B}(r)(j\omega)\, e^{j\omega t}$$

$$\vec{\nabla}\times\vec{E}(r) = -j\omega\vec{B}(r)$$

Time-harmonic Maxwell' equation. A function of space Only.

Time-harmonic Maxwell's Equations

Following the same procedure, the TH Maxell's equations are written as,

$$\vec{\nabla}\times\vec{E}(r) = -j\omega\vec{B}(r)$$

$$\vec{\nabla}\times\vec{H}(r) = \sigma\vec{E}(r) + j\omega\vec{D}(r)$$

$$\vec{\nabla}\cdot\vec{D}(r) = \rho_v(r)$$

$$\vec{\nabla}\cdot\vec{B}(r) = 0$$

Any time-harmonic electromagnetic field must satisfy the above equations.

Q & A

Send me an email if you have any questions.